

Project 1

Support Vector Machines and a Comparative Study of Classifiers

Due Date: *Monday, November 9 (11.59PM)*

Submission: on Gradescope

Instructions

The project may be completed in groups of *up to 4 students*. If you do so, please make sure that you include everyone's full name, and that you also *select everyone's name when submitting the assignment on Gradescope*. This ensures that each group member will get a grade assigned and have access to the comments from the graders.

For this assignment, the following items need to be submitted:

1. A **single PDF** that contains your code, all results, plots, and written answers (generated from R Markdown, Jupyter, etc.).
2. The **source file** used to produce the PDF (.Rmd, ipynb, ...).

This is a **project**, *not a coding assignment*. You will not be told which seed to use, which two variables to plot, or which kernel and grid to try. You **will** be expected to make those choices, document them, and defend them with the ideas from class: margin and C , kernels, hinge versus likelihood, discriminant analysis versus a probability model, and what a held-out error rate can and cannot tell you. Results in R and Python need not match; that is expected.

Data Description & Scientific Question

The data are Kepler Objects of Interest (KOIs) from the NASA Exoplanet Archive cumulative table.¹ Each row is a transit-like signal around a Kepler star. The archive labels the object as **CONFIRMED** (published planet), **FALSE POSITIVE** (the transit is not a planet), or **CANDIDATE** (still under review).

¹NASA Exoplanet Archive, cumulative KOI table: <https://exoplanetarchive.ipac.caltech.edu>

Download `kepler_koi.csv` and the codebook from Coursera. In the posted file, rows with label `CANDIDATE` have already been dropped. The file has $n = 7587$ objects (2748 `CONFIRMED`, 4839 `FALSE POSITIVE`).

Response. The response is the two-level factor

$$y = \text{koi_disposition} \in \{\text{CONFIRMED}, \text{FALSE POSITIVE}\}.$$

Predictors. Use only the **physical transit** and **stellar measurements** in the posted file. The column names are those in the codebook, including `koi_period`, `koi_duration`, `koi_depth`, `koi_impact`, `koi_model_snr`, `koi_prad`, `koi_teq`, `koi_insol`, `koi_steff`, `koi_slogg`, `koi_srad`, `koi_kepmag`, and the other physical columns in the CSV. Units differ by orders of magnitude; **that is part of the problem.**

Columns you may not use, even if you download a wider table yourself:

- any official false-positive *flag* (`koi_fpflag_*`);
- the Kepler pipeline score or disposition comment (`koi_score`, `koi_pdisposition`, `koi_comment`, and similar);
- names, identifiers, or sky coordinates used as features (`kepoi_name` is an ID only).

Those variables are how the archive recorded the label. Putting them in X turns the project into a copy of the pipeline, not a classifier.

Scientific Question

The scientific question is whether the transit geometry and the host-star properties, by themselves, distinguish confirmed planets from false positives, and which classifier from the course you would trust for that triage.

Design a fair comparison of classifiers for this two-class problem. The comparison has to be something you could **defend to an astronomer** who will spend telescope time on the objects you call *planets*.

Methods. Fit *all* of the following on the same training data, and evaluate them on the same held-out test data:

- (i) LDA;

- (ii) logistic regression;
- (iii) a linear SVM;
- (iv) a kernel SVM (you choose the kernel);
- (v) *one additional method* from the course that is not an SVM and not LDA/logistic. Reasonable choices include a classification tree (or a forest), L_1 or L_2 penalized logistic regression, or QDA. Explain why you picked it.

Protocol

You choose the split and the tuning scheme, subject to these constraints:

- There must be a held-out test set that is *not* used to choose C , a kernel width, a tree depth, a penalty, or anything else.
- The split must be stratified by the two dispositions.
- *Missing values*: drop or impute using *training-set* information only; explain what you did.
- Scale features when the method uses distances or inner products, using training-set summaries only. LDA and unpenalized logistic regression in the original units are defensible; explain what you scaled and why.
- Tuning is done by cross-validation *on the training set only*. At minimum you must tune C for both SVMs and the kernel width if you use a radial kernel.
- The whole pipeline must be reproducible: report the seed, the split fractions, the CV folds, and the grids.

Default training-set frequencies are acceptable class priors. If you change priors or the relative cost of the two errors, explain.

What to Report

The report should contain enough for a reader to reconstruct what you did and to judge the scientific claim. *At a minimum*:

- (a) A short description of the protocol (split, missing values, scaling, kernel, grids). *Do not paste pages of raw software output!!*

- (b) A comparison table with training and test error (or accuracy) for every method. Confusion matrices on the *test* set, with rows = true disposition and columns = predicted disposition, labeled by name rather than 0/1. Report the two error types separately: confirmed planets called false positives, and false positives called planets. *Overall accuracy is not enough!*
- (c) At least one figure that is not a confusion matrix: period versus planet radius by class, a CV curve in C , support-vector counts versus C , a calibration plot of logistic probabilities versus SVM scores; you choose. The caption should state what the figure is evidence *for*.
- (d) A written discussion, not a bullet paraphrase of the table. Address *all* of the following, with reference to the numbers and figure(s) you produced:
- Which method would you hand an observer who has to decide which KOIs get follow-up spectra, and why? Test accuracy alone is not a complete answer. Which error is worse in this application, and did your ranking of methods change when you looked at the two error types separately?
 - Linear SVM versus logistic regression: do they disagree on many test objects? Relate this to hinge loss versus logistic loss (points beyond the margin; a probability model versus a score).
 - Does the kernel SVM improve on the linear SVM enough to justify the extra tuning? Does a very large C look like overfit?
 - SVM scores are not $\hat{P}(\text{CONFIRMED} \mid x)$. If the observer wanted a “needs more data” bin instead of a hard call, which of your methods is actually built for that? Say what you would do instead of treating $\text{sign}(f(x))$ as a posterior.
 - One limitation of the study (missingness that is not at random, confirmed planets being easier targets, a single split, residual label leakage if you slipped a flag into X , ...) and how it should change how strongly you believe the ranking of methods.

Grading

Roughly equal weight on:

1. a correct and reproducible pipeline, including the column restrictions above
2. a comparison that includes both SVMs and the earlier classifiers, with tuning done on training data only;
3. figures and tables that a reader can interpret without guessing, including both kinds of error;

4. a discussion that uses the language of the course rather than “model A had a slightly higher accuracy.” A summary of test accuracies with no argument is not a completed project.

Remark: A tenth of a percentage point in test accuracy is not a finding. Discuss the *pattern* of errors and whether the method is doing something you would trust on the next KOI delivery.